Peiyu (Georgia) Li

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Education

University of Notre Dame, Notre Dame, IN — Aug. 2024 - Present
Master of Computer Science and Engineering   GPA: 4.0/4.0
Graduate Teaching Assistant: Elements of Computing
Advisor: *Prof. Nitesh V. Chawla (ACM, IEEE, AAAI Fellow, Frank M. Freimann Professor of Computer Science and Engineering)*
Coursework: Machine Learning, Large Language Models: Basics and Practice, Graph Learning

Sun Yat-sen University, Guangzhou, China — Aug. 2019 - Jun. 2023
Bachelor of Philosophy, Major in Logic   GPA: 3.9/4.0, Ranking: Top 4
Scholarship: Outstanding Student Scholarship of Sun Yat-sen University (2020 & 2021 & 2022) (10%)

Research Experience

Multimodal Foundation Model Integrating Recipe and Food Image Generation — Nov. 2023 – Feb.2024
Supervisor: Prof. Nitesh V. Chawla
Role: First Author — https://doi.org/10.1145/3627673.3679885

- Presented at the Conference on Information and Knowledge Management (**CIKM**) 2024, Boise, Idaho
- Developed a novel **multimodal** architecture to integrate five multimodal tasks—t2t, t2i, i2t, it2t, and t2ti—bridging text, recipe, and image generation.
- Leveraged large language models (LLMs) alongside pre-trained image encoders and decoders to enhance food computing.
- Conducted comprehensive **data preprocessing** on the Recipe 1M and Recipe 1M+ datasets (totaling **3TB**), ensuring data quality and integrity for model training and evaluation.
- Utilized CLIP ViT-L cosine similarity, BLEU score, and ROUGE score to assess food image and recipe generation, surpassing the benchmarks.

Projects

IEEE Computer Society North America Student Challenge - User Preference Inference — October 2024
Role: 1st Place Winner

- **Achieved 1st place** with a top score of **96.333%** in a competition focused on inferring user preferences from conversations between virtual users and large language models (LLMs).
- **Fine-tuned** GPT-4o-mini to optimize model performance for real-world applications in personalized AI, demonstrating expertise in natural language processing (NLP) and model adaptation.
- Utilized KAIST AI's Multifaceted Collection dataset to structure training (**2,046** samples), validation (**1,091** samples), and final test (**1,000** samples) datasets for comprehensive model evaluation.
- As 1st place winner, will present this work at **IEEE Big Data** in Washington, DC in December 2024.

Skills

Computer Skills: Java, Python, C++, Linux, JavaScript, LaTeX, Tableau, MySQL
Libraries: PyTorch, Pandas, NumPy, Matplotlib, TensorBoard
Language Skills: English (fluent speaker), Mandarin (native speaker), Cantonese (fluent speaker), German (Goethe-Zertifikat A2)